

220B Discussion: Minimax via Game Theory

Problem 1: A 2 x 2 game

Two-person zero-sum game with $\Theta = \{\theta_1, \theta_2\}$ for Nature (Player I), $\mathcal{A} = \{a_1, a_2\}$ for the statistician (Player II), and loss matrix (rows indexed by θ , columns by a):

$$L = \begin{pmatrix} 0 & 4 \\ 3 & 1 \end{pmatrix}.$$

- (a) Compute $\underline{V} = \max_{\theta} \min_a L(\theta, a)$ and $\overline{V} = \min_a \max_{\theta} L(\theta, a)$. Show that $\underline{V} < \overline{V}$ (no pure-strategy value).
- (b) Let $\delta^* = (q_1, q_2)$ for Player II (so $q_1 + q_2 = 1$). Solve $L(\theta_1, \delta^*) = L(\theta_2, \delta^*)$ for (q_1, q_2) and report the common value.
- (c) Let $\pi^* = (p_1, p_2)$ for Player I. Solve $L(\pi^*, a_1) = L(\pi^*, a_2)$ for (p_1, p_2) and report the common value.
- (d) Verify the hypothesis of Theorem 2 of Lecture 6: check $L(\theta_i, \delta^*) \leq L(\pi^*, a_j)$ for all four pairs (i, j) . Conclude that the game has value V and identify it.
- (e) Apply Theorem 4 to compute Θ_V . Use Theorem 1 to argue that π^* found in (c) is the unique maximin prior.

Problem 2: Continuous action, three states for Nature

Two-person zero-sum game with $\Theta = \{1, 2, 3\}$, $\mathcal{A} = \mathbb{R}$, $L(\theta, a) = (\theta - a)^2$. No data.

(a) For any prior π on Θ , show

$$\inf_{a \in \mathbb{R}} L(\pi, a) = \text{Var}_{\pi}(\theta), \quad \text{attained at } a = \mathbb{E}_{\pi}[\theta].$$

(b) Find the maximin prior $\pi^* = \arg \max_{\pi} \text{Var}_{\pi}(\theta)$ over distributions on $\{1, 2, 3\}$ and report $\underline{V}$.

(c) Set $\delta^* = a^* = \mathbb{E}_{\pi^*}[\theta]$. Compute $L(\pi^*, a)$ as a function of $a \in \mathbb{R}$, and $L(\theta, a^*)$ for $\theta \in \{1, 2, 3\}$.

(d) Show $\sup_{\theta} L(\theta, a^*) \leq \inf_{a \in \mathbb{R}} L(\pi^*, a)$ and conclude that a^* is a minimax strategy with $V = \underline{V}$.

(e) Show that no $a \in \mathbb{R}$ is an equalizer in this problem. Identify the set $\{\theta : L(\theta, a^*) = V\}$ and check that π^* is supported on it.